Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/30/2006
PI Name:	Wiener, Michael Ph.D.		
Project Title:	A Directed Approach to Membrane Protein Crystallography		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Protein crystal growth		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
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No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
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The structural determination of integral membrane proteins is a frontier area of modern structural biology. In contrast to soluble proteins, exceedingly few integral membrane protein structures have been solved by x-ray crystallography. This paucity has little to do with crystallography per se, but rather with two substantial obstacles that precede structure determination. First, production of sufficient quantities of integral membrane proteins, particularly eucaryotic proteins, for crystallization is not generally successful. Second, purified integral membrane proteins, in the form of protein-detergent complexes (PDCs), have historically been very difficult to crystallize in forms suitable for structure determination by x-ray diffraction. This proposal addresses these two challenges with a directed approach for membrane protein crystallography. This approach is based upon two hypotheses: the bacterial expression of bacterial and archaebacterial homologs of eucaryotic membrane proteins is a tractable approach to generating sufficient quantities of protein for crystallization experiments; and membrane protein crystallization screens, based upon physical-chemical properties of detergent and protein-detergent solutions, will effectively yield crystals or leads for crystals. Objective 1:

Task Description:	Select an array of target eucaryotic proteins that have bacterial & archaebacterial homologs, and conduct cloning and expression tests of these homologs. Objective 2: Purify homologs and conduct crystallization experiments that utilize novel screens being developed in this laboratory. Objective 3: Obtain diffraction-quality crystals of membrane proteins and proceed with structure determination. Membrane protein structure determination is a frontier area of modern structural biology. Twenty to thirty-five percent of the proteins encoded by an organism encode membrane proteins. The majority of drugs are directed towards membrane protein targets. Structures of membrane proteins obtained by x-ray crystallography of well-ordered three-dimensional crystals can provide key insight into mechanism, and can serve as input for computational and structure-based drug design efforts. Despite both the obvious fundamental and applied importance of membrane protein structures, few have been solved. The proposed research aims to develop methods and apply them to the crystallization of various membrane proteins. The development and success of these methods will help to make this area of structural biology less technically daunting, and holds the promise of enabling in the future the more expeditious structural determination of a wide range of membrane proteins.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)